



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 56: Openroad VLSI Design 2

Status Update 2

Kevin, Talha, Githin, Giovani

Sponsor: Kevin Nowka

TA: Michael Johnson-Moore

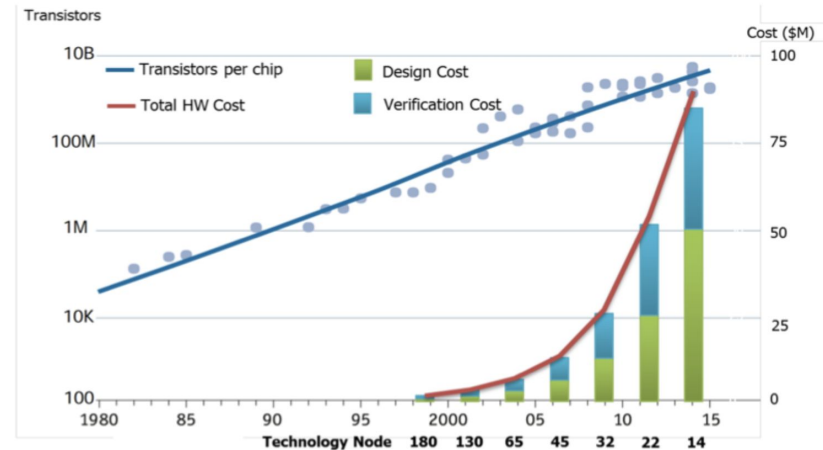
Project Summary

Problem statement:

- Traditional RTL-to-GDSII chip design is expensive, creating a cost barrier to developing a System on a Chip (SoC).

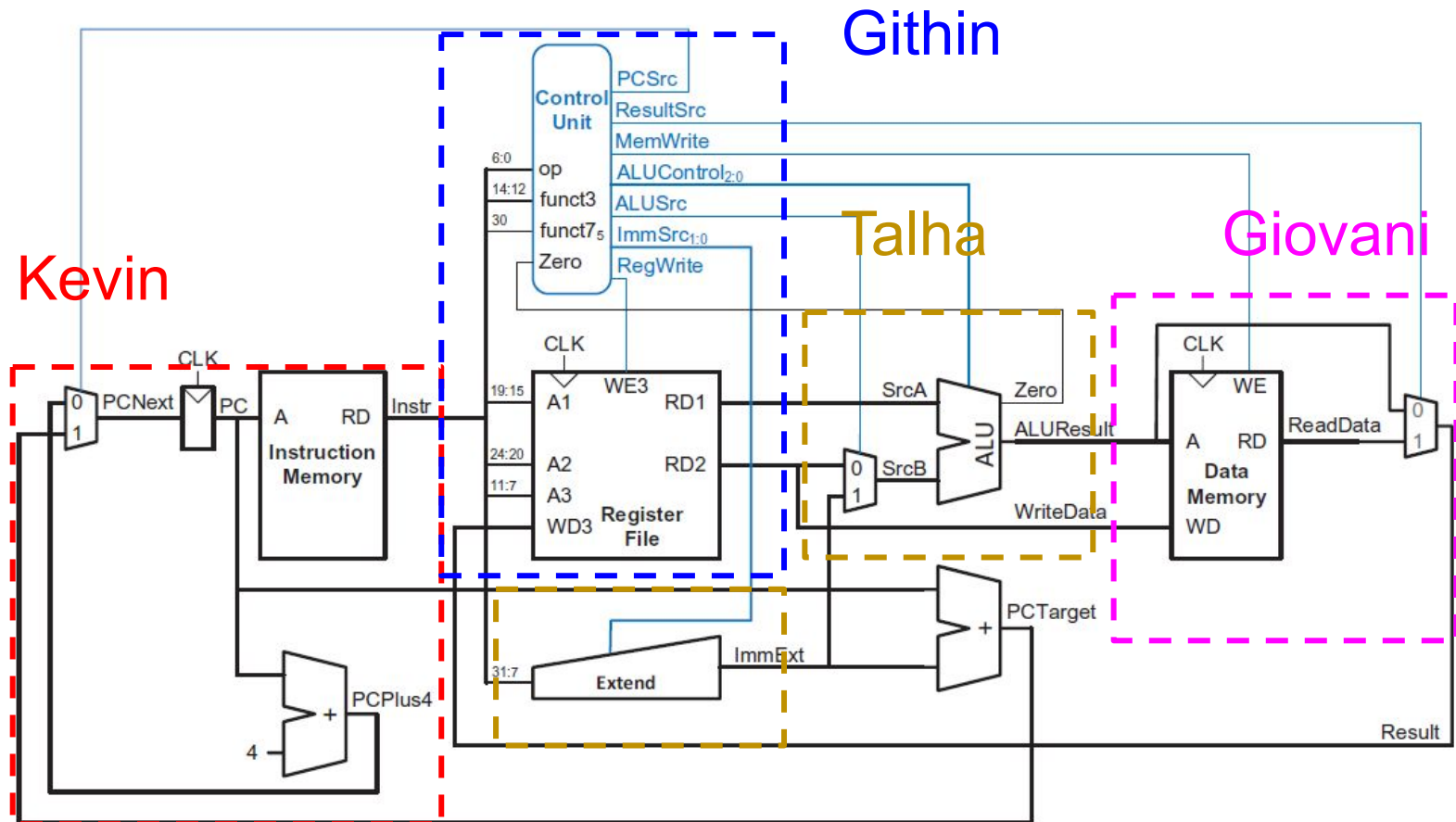
Our Solution:

- Create a 16-bit RISC-V using OpenROAD, which automates chip design by only needing only an RTL description to generate GDSII, significantly reducing cost.



OpenROAD/OpenLane

System Overview





Project Timeline

Finalize individual subsystem (completed 1/26)	Start integration/validation (Start by 2/2)	Implement pipeline stages (Start by 2/16)	Validate pipeline stages (Start by 3/2)	Start OpenLane/OpenROAD process (Start by 3/23)	Final Validation (Start by 4/6)
--	---	---	---	---	---------------------------------

Fetch

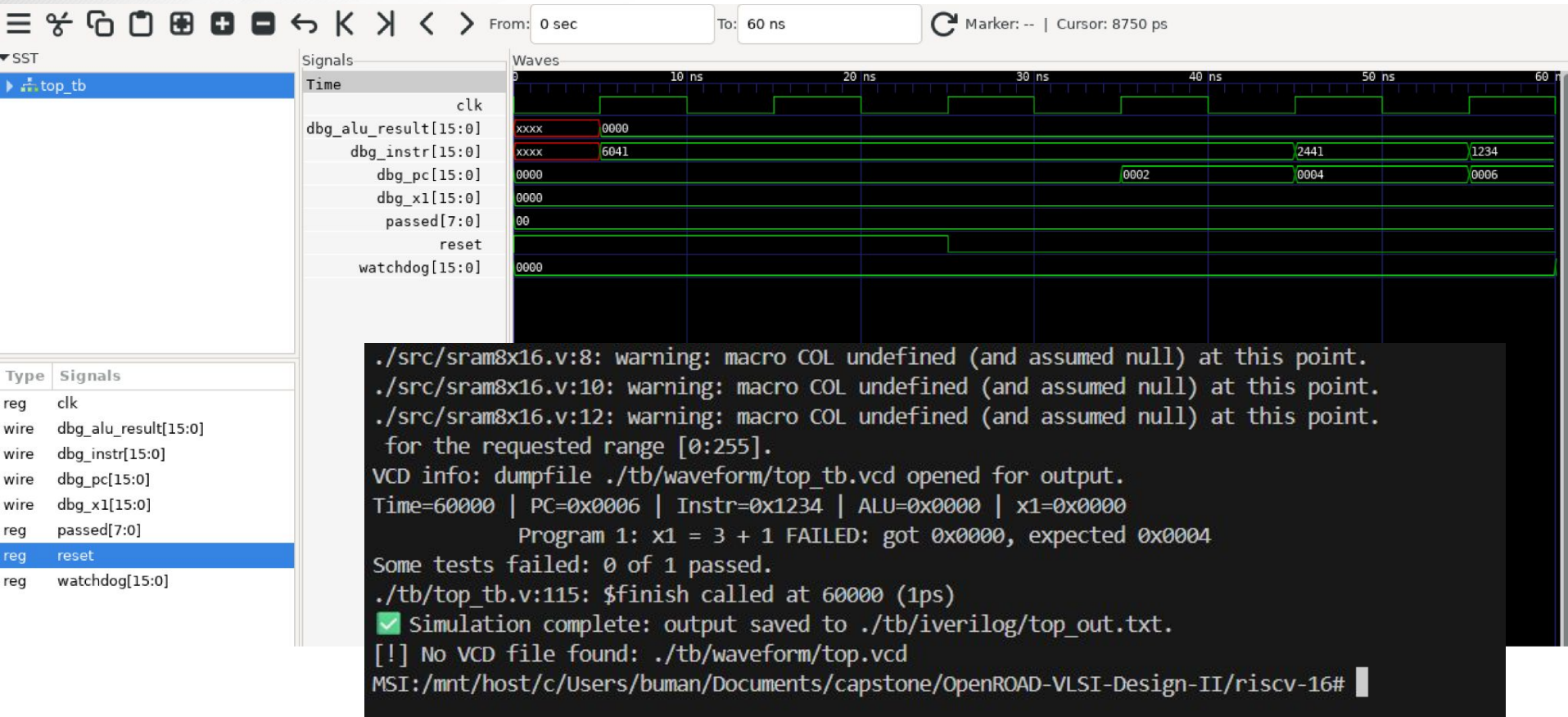
Kevin Umanzor

Accomplishments since status update 1 12 hrs of effort	Ongoing progress/problems and plans until next presentation
• Worked on creating the full-system CPU testbench and added basic initial test cases to support early bring-up and debugging. • Decided to proceed with Tiny Tapeout as the fabrication path for the project	• Finish the full-system testbench by adding multiple instruction/program test cases and ensuring all tests pass consistently.



Fetch

Kevin Umanzor



Decode

Githin Johny

Accomplishments since status update 1 12 hrs of effort	Ongoing progress/problems and plans until next presentation
Completed top-level implementation with Talha	- Debug single-cycle testing of processor - Monitor internal signals for waveform viewing

Top-level module

Githin Johnny

```
module riscv16_top (
    input wire      clk,
    input wire      reset,

    // Debug outputs
    output wire [15:0] dbg_pc,
    output wire [15:0] dbg_instr,
    output wire [15:0] dbg_alu_result,
    output wire [15:0] dbg_x1
);
```

```
/* =====
   Debug
   ===== */
assign dbg_pc          = PC;
assign dbg_instr       = instruction;
assign dbg_alu_result  = ALUResult;
assign dbg_x1          = rf_dbg_x1;
```




Execute

Talha Ibrahim

Accomplishments since status update 1 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Implemented the Single Cycle processor top level design with Githin	• Debugging code for various test cases • Research pipeline implementation for chip

Top-level Module

Talha Ibrahim

```

/* =====
|   Instruction Fields
|   ===== */
wire [2:0] opcode = instruction[2:0];
wire [3:0] func   = instruction[6:3];
wire [2:0] rs1    = instruction[15:13];
wire [2:0] rs2    = instruction[12:10];
wire [2:0] rd     = instruction[9:7];

/* =====
|   Control Unit
|   ===== */
wire      PCSrc;
wire      ResultSrc;
wire      MemWrite;
wire      ALUSrc;
wire      RegWrite;
wire [1:0] ImmSrc;
wire [3:0] ALUControl;
wire      zero;

```

```

// Control unit
Control_unit CU (
    .opcode(opcode),
    .func(func),
    .zero(zero),
    .PCSrc(PCSrc),
    .ResultSrc(ResultSrc),
    .MemWrite(MemWrite),
    .ALUControl(ALUControl),
    .ALUSrc(ALUSrc),
    .ImmSrc(ImmSrc),
    .RegWrite(RegWrite)
);

/* =====
|   Register File
|   ===== */
wire [15:0] RD1, RD2, WD3;
wire [15:0] rf_dbg_x1;

```



Data Memory

Giovani Giagnacovo

Accomplishments since status update 1 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Debugged incompatibility issues between top level module and data memory module.	• Further Debug SRAM and Writeback MUX to ensure functionality in top level module (entire CPU) • Research pipelining by using 64 bit CPU SRAM example.



Data Memory

Giovani Giagnacovo

```
src\sram8x16.v:7: warning: macro COL undefined (and assumed null) at this point.
src\sram8x16.v:9: warning: macro COL undefined (and assumed null) at this point.
src\sram8x16.v:11: warning: macro COL undefined (and assumed null) at this point.
src\sram8x16.v:11: warning: macro ROW_D undefined (and assumed null) at this point.
tb\op_tb.v: No such file or directory
src\top.v:44: warning: parameter MEMFILE not found in riscv16_top.IMEM.
src\top.v:73: error: Unknown module type: Control_Unit
src\top.v:141: error: Unknown module type: Data_Memory
4 error(s) during elaboration.
*** These modules were missing:
    Control_Unit referenced 1 times.
    Data_Memory referenced 1 times.
***
```



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Parts Ordering Status

Not Applicable

Execution & Plan

[illegible]



Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.1	Tile Size Constraint	The final GDSII layout must remain within the 2-tile maximum area allowed for TinyTapeout.	Run final Place and Route in OpenROAD, verify bounding box and padframe placement using the layout area report.	Untested	Full Team
4.1	DRC	Zero DRC violations (including antenna) in Magic and KLayout using the Sky130A/TinyTapeout rule decks.	Run Magic and KLayout DRC on the final GDS and iterate fixes until zero violations.	Untested	Full Team
4.2	Operating Temperature Range	Design must remain functionally valid across process and temperature corners (-55°C to +125°C).	Run timing and power analysis using PDK corner simulations (TT, SS, FF) to ensure performance remains stable within range.	Untested	Talha Ibrahim, Kevin Umanzor
5.1	LVS	Netgen reports "LVS clean" with 0 opens/shorts and all devices/nets/pins matched.	Extract the layout to SPICE and compare against the synthesized gate-level netlist in Netgen with the sky130A setup.	Untested	Full Team
5.2	Primary Input Power	Active power below 200 μ W and lower during idle conditions.	Estimate and verify power through post-layout power analysis and simulation reports in OpenROAD.	Untested	Talha Ibrahim, Kevin Umanzor
5.3	Signal Interfaces	All interfaces (I ² C, SPI, USB) respond correctly at specified logic levels and timings.	Run RTL and gate-level testbenches to verify correct data transfer, logic levels, and signal timing for all interfaces.	Untested	Githin Johny, Giovanni Giagnacovo
5.4	User Control Interface	The reset mechanism returns the CPU to a known initial state.	Run RTL simulations for both synchronous and asynchronous resets to confirm the PC and register file reset correctly.	Untested	Giovanni Giagnacovo, Talha Ibrahim
5.5	Clock Interface	The design accepts a 10 MHz external clock and meets all setup and hold timing constraints.	Run Static Timing Analysis (STA) and post-layout timing simulations in OpenROAD to verify clock stability and timing closure.	Untested	Kevin Umanzor, Githin Johny
6.1	Fetch	The Fetch stage retrieves correct instructions and updates the program counter properly.	Create directed and random test programs in RTL simulation to verify instruction fetch and PC increment behavior.	Tested	Kevin Umanzor
6.2	Decode	The Decode stage correctly generates control signals from instruction fields.	Run RTL testbenches for all supported instructions to confirm control signal accuracy and complete coverage.	Tested	Githin Johny
6.3	Execute	The ALU produces correct results for all arithmetic and logic operations.	Run RTL and gate-level simulations for arithmetic, logical, and branch operations with directed and random test vectors.	Tested	Talha Ibrahim
6.4	Memory/Write Back	The memory and write-back stages perform correct data reads, writes, and register updates.	Run RTL and gate-level simulations with memory models to verify correct load/store behavior and data integrity.	Tested	Giovanni Giagnacovo
	Full System Simulation	The full 16-bit RISC-V design executes test programs correctly through all pipeline stages.	Integrate all modules into a top-level RTL and post-layout simulation, verifying instruction execution, output correctness, and timing.	Untested	Full Team
	Full System Demo	Execute a test program on each corresponding part on the chip measuring correct output for all cases; along with measured power and basic I/O pass	Integrate tests for top level regression and postlayout simulation capturing outputs.	Untested	Full Team



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Thank you!!